

Non-Autonomous System

BSA, 2024

```
In[1]:= ClearAll["Global`*"]
```

Initial Data № 7.4

```
In[2]:= X = {x[1][t], x[2][t], x[3][t]};  
P = {{a, Sin[2 * t], 0}, {Cos[2 * t], -a, -Sin[t]}, {0, Cos[t], -1}};  
Eq = Thread[D[X, t] == P . X];
```

```
In[5]:= Print[TableForm[Eq] /. {x[i_] => Subscript[x, i]}]
```

$$\begin{aligned}x_1'[t] &= a x_1[t] + \sin[2t] x_2[t] \\x_2'[t] &= \cos[2t] x_1[t] - a x_2[t] - \sin[t] x_3[t] \\x_3'[t] &= \cos[t] x_2[t] - x_3[t]\end{aligned}$$

```
In[6]:= Print[MatrixForm[D[X, t] /. {x[i_] => Subscript[x, i]}], " = ", MatrixForm[P], " x ", MatrixForm[X /. {x[i_] => Subscript[x, i]}]]
```

$$\begin{pmatrix} x_1'[t] \\ x_2'[t] \\ x_3'[t] \end{pmatrix} = \begin{pmatrix} a & \sin[2t] & 0 \\ \cos[2t] & -a & -\sin[t] \\ 0 & \cos[t] & -1 \end{pmatrix} \times \begin{pmatrix} x_1[t] \\ x_2[t] \\ x_3[t] \end{pmatrix}$$

Compute Eigenvalues

```
In[7]:= T = 2 * Pi;
```

```
In[8]:= NA = Transpose[X /. Table[NDSolve[Union[Eq /. a → #, Thread[(X /. t → 0) == UnitVector[3, i]]], X, {t, 0, T}][[1]], {i, 3}]] /. {t → T} &;
NI = Eigenvalues[NA[#]] &;
NIAbs = Abs[NI[#]] &;
```

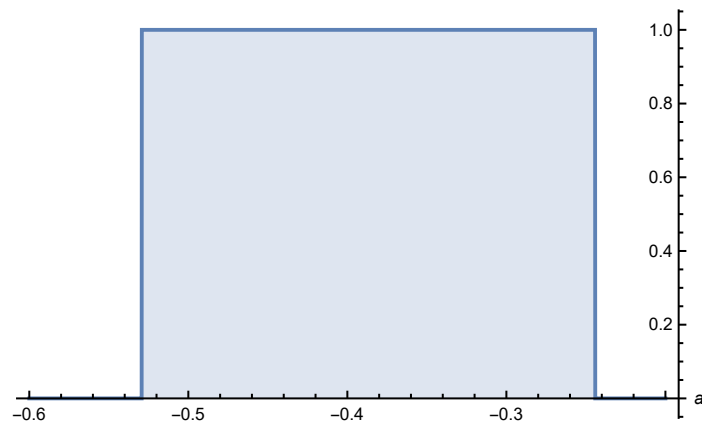
```
In[11]:= LExp = N[Exp[Integrate[Tr[P] /. {a → #}, {t, 0, T}]]] &;
LI =  $\rho$  /. NSolve[ $\rho^3 + \text{LExp}[\#] * (-\rho^2 + \rho - 1)$ ,  $\rho$ ] &;
LIAbs = Abs[LI[#]] &;
```

```
In[14]:= NExp = Apply[Times, NIAbs[#]] &;
```

Stability Domain

```
In[15]:= Plot[If[Max[NIAbs[a]] < 1, 1, 0], {a, -0.6, -0.2}, AxesLabel → {a, None}, Filling → Axis, FillingStyle → Automatic]
```

Out[15]=



Left Border

In[16]:= `Round[NIAbs[-0.5292], 0.001]`

Out[16]=
`{1., 0.15, 0.012}`

Right Border

In[17]:= `Round[NIAbs[-0.2442], 0.001]`

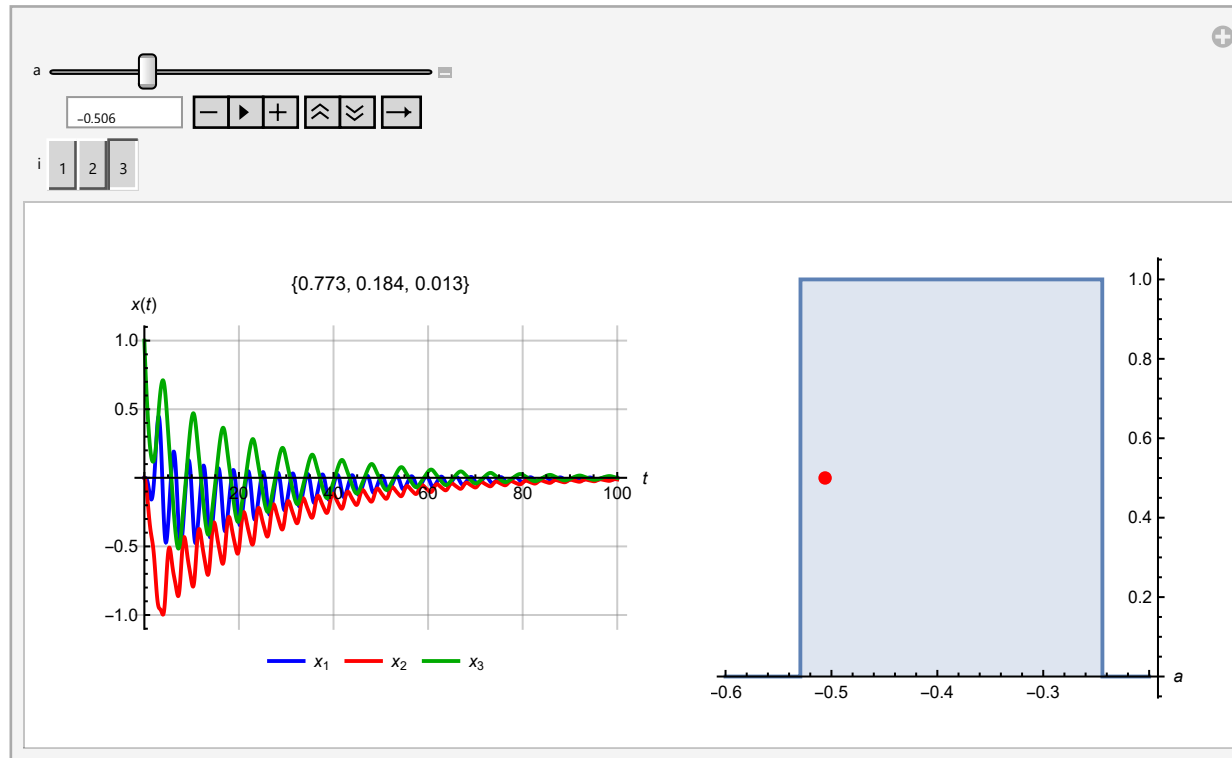
Out[17]=
`{1., 0.085, 0.022}`

Plot the Solution

```
In[18]:= PlotSolution[α_, τ_, i_] := (
  FX = X /. NDSolve[Union[Eq /. a → α, Thread[(X /. t → 0) == UnitVector[3, i]]], X, {t, 0, τ}][[1]];
  p1 = Plot[FX, {t, 0, τ}, PlotRange → Full, PlotLegends → Placed[{Subscript[x, 1], Subscript[x, 2], Subscript[x, 3]}, Bottom],
    AxesLabel → {t, x[t]}, GridLines → Automatic, PlotStyle → {Blue, Red, Darker[Green]},
    PlotLabel → Style[Round[NIAbs[α], 0.001], 10], ImageSize → Scaled[0.9]];
  p2 = Show[Plot[If[Max[NIAbs[a]] < 1, 1, 0], {a, -0.6, -0.2}, AxesLabel → {a, None}, Filling → Axis, AspectRatio → 1],
    Graphics[{Red, PointSize[Large], Point[{α, 0.5}]}], ImageSize → Scaled[0.7]];
  GraphicsRow[{p1, p2}]
);
```

```
In[19]:= Manipulate[PlotSolution[a, 100, i], {a, -0.6, -0.2, Appearance -> "Open"}, {i, {1, 2, 3}}]
```

Out[19]=



Check with Liouville Formula

```
In[20]:= Plot[{NExp[a], LExp[a]}, {a, -0.6, -0.2}, PlotRange → Full, PlotLegends → Placed[{"Numerical", "Liouville Formula"}, Bottom],
  AxesLabel → {a, Subscript[ρ, 1] * Subscript[ρ, 2] * Subscript[ρ, 3]}, GridLines → Automatic, PlotStyle → {Blue, Red}]
```

Out[20]=

